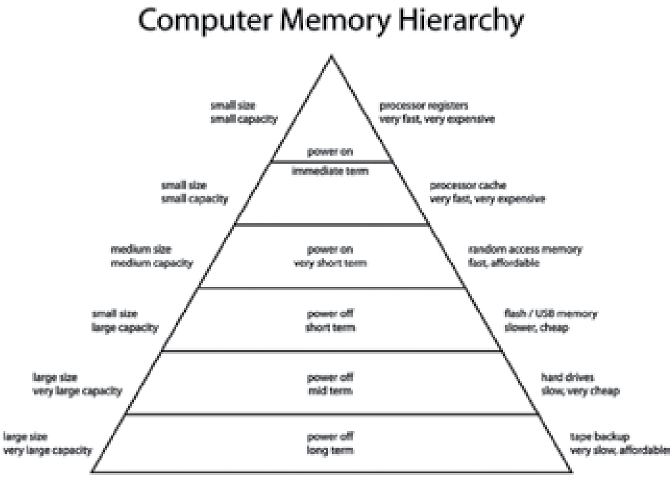

to be static RAM which could store data for longer periods and then cam DDR RAM which increased the rate at which data transfer was taking place. The evolution happened in this order - SDRAM > RDRAM > DDR > DDR2 > DDR3. With each passing generation the improvements also included latency, speed and reduced power consumption. Physically they have been made to look a bit different so that they cannot be used interchangeably between different generations. This is achieved with a notch in the contact pins and each generation of DDR memory has a different offset as shown in the next page.

A few specialized RAM types exist :-

1. ECC - (Error Checking and Correcting) memory constantly checks the data received and verifies it's parity. This is a way of counting the 1's and 0's and checking whether an extra bit has sneaked in a particular data stream or not. Such memory are usually preferred by servers since a small mistake creeping in one input could change the results completely.

2. Buffered - Another server memory is where an intermediate stage exists with a small CPU register that holds data which cannot be processed by the RAM presently, so the data is stored for a short while.

Parameters / Specification

1. Speed - Memory speed depends on operating frequency and bandwidth. An easy way to calculate is by multiplying the frequency with the number